

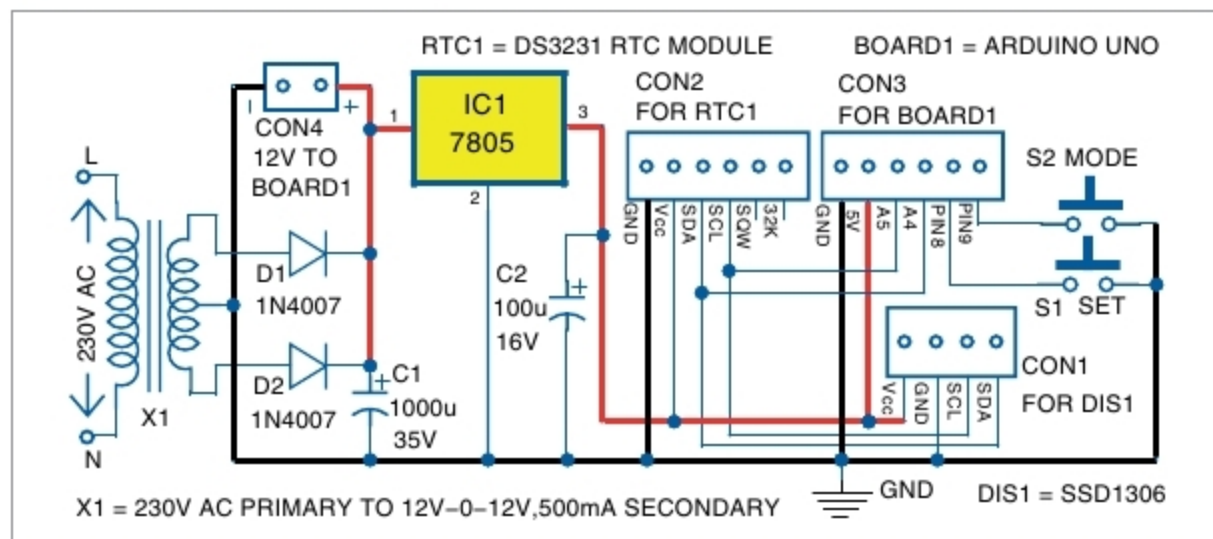


Fig. 3: Circuit diagram of Arduino based real-time clock (Pins A4, A5, 5V of Board1)

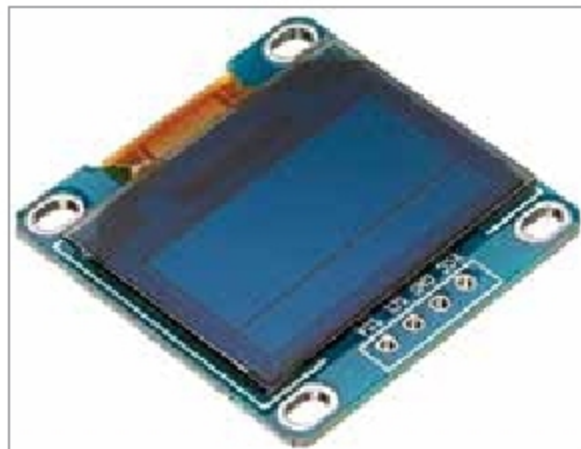


Fig. 4: SSD1306 OLED display module



Fig. 5: DS3231 RTC module

compensation for leap years for and months with less than 31 days.

The module can work on either 3.3V or 5V, which makes it suitable for many development platforms or microcontrollers. The battery input is 3V, and a typical CR2032 3V battery can power the module and maintain the information for more than a year. The DS3231 module (front and back) used in this project is shown in Fig. 5.

The I2C communication protocol used in the module makes connections to the Arduino board very easy. All we need are four pins: VCC and GND pins for powering the module, and SDA and SCL pins for I2C communication.

PARTS LIST	
<i>Semiconductors:</i>	
IC1	- 7805, 5V voltage regulator
Board1	- Arduino Uno
RTC1	- DS3231 RTC module
DIS1	- SSD1306 OLED module
D1, D2	- 1N4007 rectifier diode
<i>Capacitors:</i>	
C1	- 1000µF, 35V electrolytic
C2	- 100µF, 16V electrolytic
<i>Miscellaneous:</i>	
X1	- 230V AC primary to 12V-0-12V, 500mA secondary transformer
S1, S2	- Push-to-on switch
CON1	- 4-pin male and female Berg strip
CON2, CON3	- 6-pin male and female Berg strip
CON4	- Male DC power jack
	- Jumper wires
	- Heatsink for IC1

Power supply. The regulated power supply is designed with a full-wave rectifier built around 1N4007 diodes (D1 and D2), two capacitors (C1 and C2), 5V voltage regulator 7805 (IC1) and 230V AC primary to 12V-0-12V, 500mA secondary step-down transformer (X1). In the project, two power supplies are required: 12V DC for Arduino Uno power jack, and 5V for DS3231 RTC and SSD1306 OLED module.

Arduino Uno. The heart of the project is the Arduino Uno R3 board, which is based on ATmega328/ATmega328P microcontroller. It has 14 digital input/output (I/O) pins, six analogue inputs, 32k flash memory, 16MHz crystal oscillator, USB connection, a power jack, an ICSP header, a reset button, among others. It can be programmed using Arduino IDE software.

In the circuit, there are two push buttons: S1 for setting and S2 for mode

EFY Note

The source code of this project is available for free download at source.efymag.com

change. These are used for setting the time and date of the real-time clock. The ground connection of Arduino needs to be connected to ground connections of RTC1 module and display DIS1 module.

Construction and testing

After making all the connections, test the working of the circuit by pressing the mode button (S2), and then release it. The cursor on display will flash on the day-of-week field. Press and hold the set button (S1) to advance the day of the week to the next day at 5Hz rate, or do a short press to advance to the next setting.

Press and release the mode button (S2) to advance the cursor to the month field. Press and hold the set button (S1) to scroll through the months quickly, or do a short press to advance to the next setting. Press and release the S2 button to advance the cursor to the year field. Press and hold the S1 button to scroll through the years quickly, or do a short press to advance to the next setting. Press and release the S2 button to advance the cursor to the hour field. Press and hold the S1 button to scroll through the hours quickly, or do a short press to advance to the next setting. The clock uses only 24-hour mode in the project. Press and release the S2 button to advance the cursor to the minutes' field. Press and hold the S1 button to scroll through the

Note that the RTC chip knows the correct number of days in each month. Press and release the S2 button to advance the cursor to the year field. Press and hold the S1 button to scroll through the years quickly, or do a short press to advance to the next setting. Valid ranges are 2000 to 2099. The RTC chip keeps track of leap years automatically. When you first power up the clock, the date will be January 1, 1900. Just advance the year to the correct value.

Press and release the S2 button to advance the cursor to the hour field. Press and hold the S1 button to scroll through the hours quickly, or do a short press to advance to the next setting. The clock uses only 24-hour mode in the project. Press and release the S2 button to advance the cursor to the minutes' field. Press and hold the S1 button to scroll through the